

PHARMACY INVENTORY & EXPIRY MANAGEMENT SYSTEM

Practical Hospital Pharmacy Operations Platform

Revised Project Workflow & Implementation Report

Field	Details
Domain	Healthcare Operations / Hospital Pharmacy
Frontend	React.js + Chart.js
Backend	FastAPI (Python)
Current database	SQLite for local deployment; PostgreSQL-ready configuration
Core capabilities	Inventory, expiry safety, FEFO dispensing, sales billing, procurement and alerts

Prepared as a revised practical workflow document for the implemented final-year project system.

1. Practical Project Scope

The system digitises the hospital pharmacy workflow from medicine catalogue creation and batch receipt to safe issue, pharmacy-counter sales, alerting, storage lookup, reporting and controlled procurement. The implementation prioritises usable daily operations while retaining forecasting support for reorder planning.

In scope in the developed system	Operational outcome
Role-based access	Admin, Pharmacist and Viewer views are separated at both UI and API levels.
Batch and expiry tracking	Each batch keeps quantity, expiry, supplier and physical storage location.
FEFO dispensing	The earliest valid-expiry batch is selected for issue or sale.
Critical stock alerts	Low stock is critical only at five or fewer total units for a medicine.
Email alerts	Configured Admin and Pharmacist recipients receive threshold-crossing notifications.
Sales and buyer billing	Pharmacists can sell medicine and issue a detailed printable bill.
Procurement review	Admin can review safety-driven supplier requests before sending them.

2. Architecture

React single-page application → JWT-authenticated FastAPI REST API → relational database. The frontend provides role-specific navigation and polished operational screens. The backend applies RBAC to every sensitive endpoint and records stock movements. Forecast calculations are served through an internal forecasting service module.

3. Role-Based Workflow

Role	Access in the developed system
Admin	Manages users and alert emails; maintains catalogue and batches; views reports, suppliers, transactions, storage, forecasts and procurement; approves supplier orders.
Pharmacist	Receives stock, manages batches, dispenses using FEFO, sells medicine, generates bills, views alerts, storage map, reports and reorder information.
Viewer	Receives a deliberately restricted, read-only safety overview. Hospital operational details, suppliers, locations, transactions, reports, forecasts and orders are not retrievable.

4. Inventory and Storage Workflow

Admin or Pharmacist creates a medicine record and records each received batch with supplier, received date, expiry date, quantity and an approved storage code. The Storage Reference screen is location-first: it separates a medicine location directory from a storage-station guide for drawers, refrigerators and controlled areas.

Stock is reduced through FEFO dispensing or sales. Each movement creates an immutable inventory transaction. Batch records retain the location code, enabling the pharmacist to find stock quickly without relying on memory.

5. Safety Alert Workflow

Near-expiry alerts are evaluated from batch expiry dates. Low-stock alerts are evaluated from the aggregate quantity of all batches for a medicine. The safety threshold is fixed at ≤ 5 units; it is intentionally separate from the planning reorder level.

When stock crosses from above five to five or fewer units, the backend sends an SMTP email to saved Admin and Pharmacist alert-email addresses. The crossing rule avoids repeated emails for the same continuing critical state.

6. Pharmacist Dispensing, Sales and Billing

The dispensing workspace accepts a SKU or batch number and finds the earliest-expiring valid batch. The pharmacist enters quantity, receiving department and optional note. The system validates available quantity, decrements stock and logs the FEFO transaction.

Sales & Billing step	System action
1. Identify item	Pharmacist enters a medicine SKU or batch number.
2. FEFO validation	System selects the earliest valid saleable batch and checks quantity.
3. Buyer entry	Buyer name, optional phone number, quantity and unit price are entered.
4. Invoice creation	A unique sale invoice is created and a Sale inventory transaction is recorded.
5. Printable bill	Bill displays invoice number, buyer details, medicine, SKU, batch, quantity, price, total, seller and date/time.
6. Safety check	If the sale crosses the critical threshold, notification email is triggered.

7. Procurement Workflow

Safety scans group low-stock and near-expiry risks into reviewable procurement requests. Requests are visible only to Admin. The Admin explicitly approves a request before it is sent to the supplier; the system records the supplier order and generates a pro-forma invoice. This preserves clinical safety while preventing automatic, unreviewed purchasing.

8. Forecasting and Reorder Planning

Historical consumption records are analysed for 30, 60 and 90-day horizons. The current application provides a forecasting service and reorder recommendation view. The next academic enhancement is to replace or extend the current estimator with validated Prophet/XGBoost models, performance metrics and scheduled retraining.

9. Reports, Audit and Data Controls

Capability	Developed behaviour
Inventory export	CSV export provides medicine, batch, quantity, expiry and reorder information.
Alert export	CSV and PDF alert reports provide critical-stock and near-expiry information.
Transaction record	Dispensing, purchase and sale movements are retained with reference values.
Audit record	Procurement and critical-stock email actions are auditable.
Sensitive data control	API role dependencies prevent Viewer accounts from obtaining protected hospital operational data.

10. Core Data Model

Entity	Purpose
users	Credentials, role and optional alert email.
medicines	Master medicine record, unit, category, reorder target and ideal stock.
batch_stocks	Batch number, supplier, quantity, dates and physical storage location.
inventory_transactions	Immutable issue, purchase, dispense and sale movement history.
sale_invoices	Buyer, item, batch, quantity, pricing, total, seller and sale timestamp.
procurement_requests / invoices	Admin-reviewed supplier order workflow.
storage_locations	Approved drawers, shelves, refrigerators and controlled zones.

11. Key API Workflows

Method	Endpoint	Role
POST	/auth/login	All authenticated users
POST	/batches	Admin, Pharmacist
POST	/dispense	Admin, Pharmacist
POST	/sales	Admin, Pharmacist
GET	/alerts	Admin, Pharmacist
POST	/procurement/scan	Admin
GET	/reports/alerts.pdf	Admin, Pharmacist

12. Implementation Status

Module	Status	Notes
Authentication and RBAC	Completed	JWT login, protected API routes and role-aware frontend navigation.
Medicine and batch management	Completed	CRUD, locations, batch tracking and expiry dates.
FEFO dispensing	Completed	Valid earliest-expiry selection and transaction logging.
Safety alerts and email	Completed	Near-expiry, ≤5 low-stock alerts and SMTP notification hook.
Sales and buyer bill	Completed	Counter sale, persisted invoice and print-friendly bill.
Procurement review	Completed	Admin-controlled safety scan, supplier order and invoice record.
Forecasting UI	Completed	30/60/90-day display and reorder guidance.
Prophet/XGBoost validation	Planned enhancement	Model comparison, metrics and scheduled retraining remain future academic work.
Automated scheduler	Planned enhancement	Nightly scans and weekly retraining can be added for production deployment.

13. Testing and Deployment

The frontend is production-built using Vite and the backend source is compilation-checked. Recommended next validation includes API tests for role boundaries, FEFO selection, stock threshold crossing, invoice creation and SMTP delivery using a test mail account. Docker Compose and PostgreSQL configuration remain available for a production-oriented deployment phase.

14. Conclusion

This revised workflow documents the system as a practical pharmacy operations platform. It preserves the academic strengths of batch-level stock safety, expiry monitoring, forecasting and RBAC, while adding controlled procurement, pharmacist sales billing, physical storage clarity and critical-email escalation for day-to-day usability.